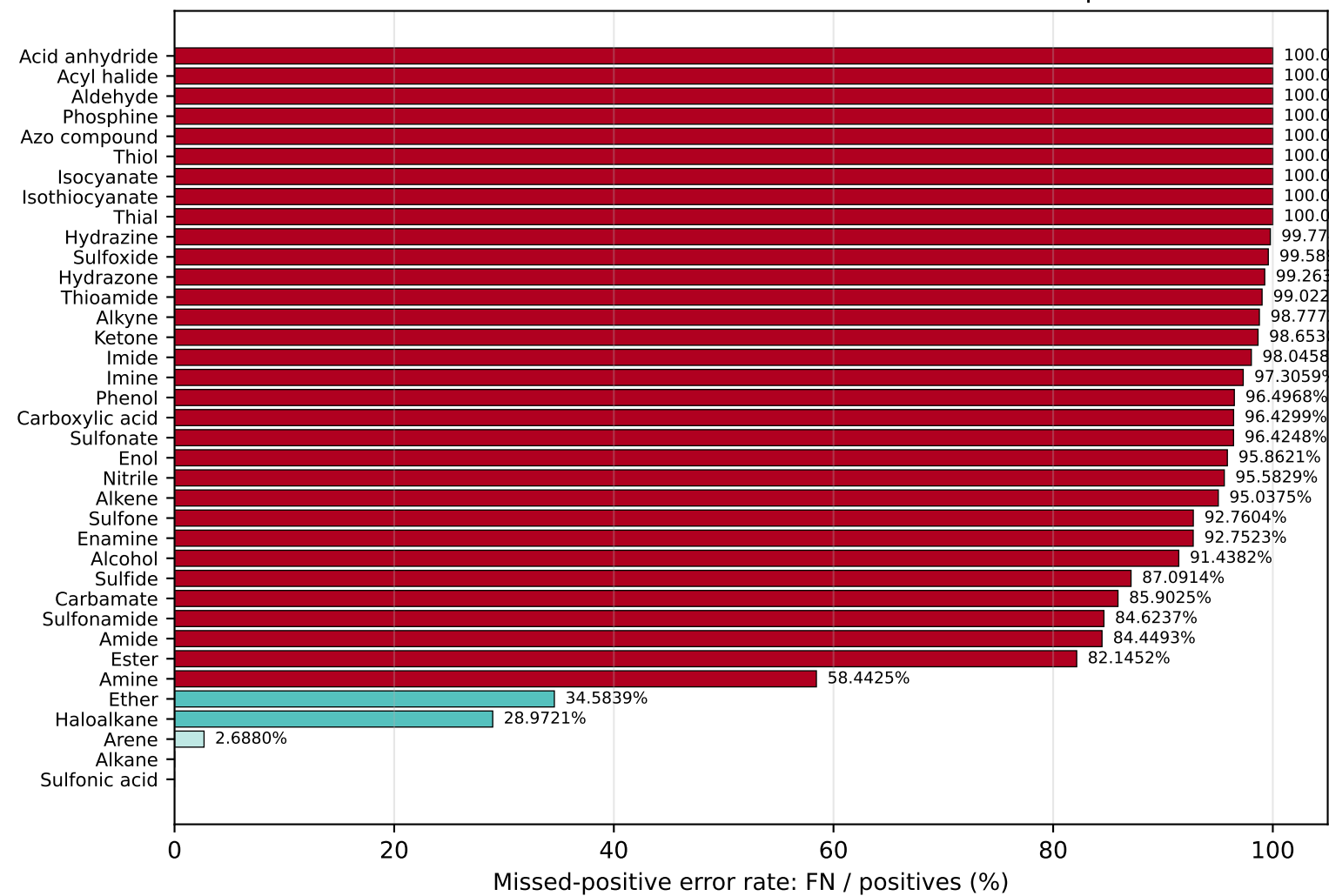
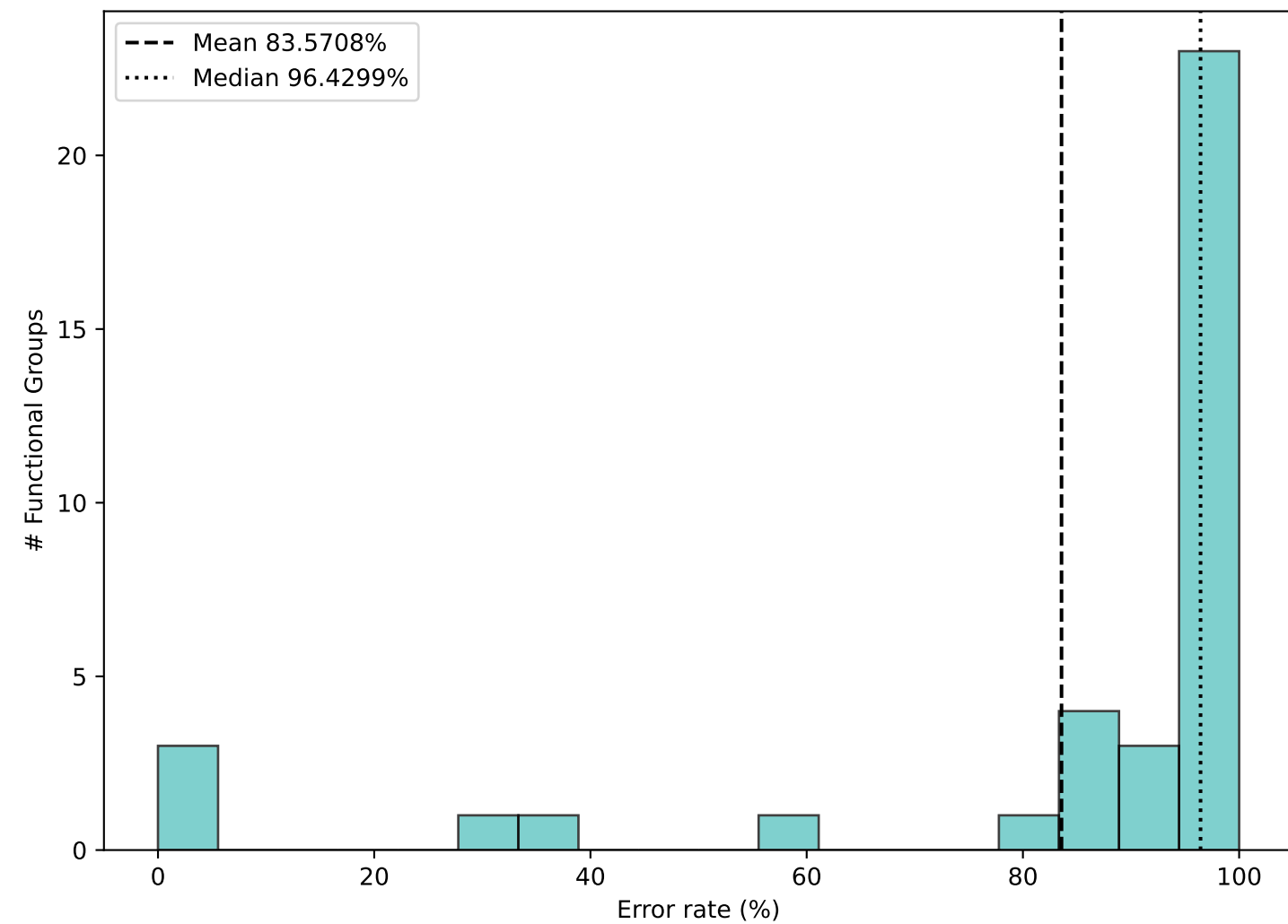


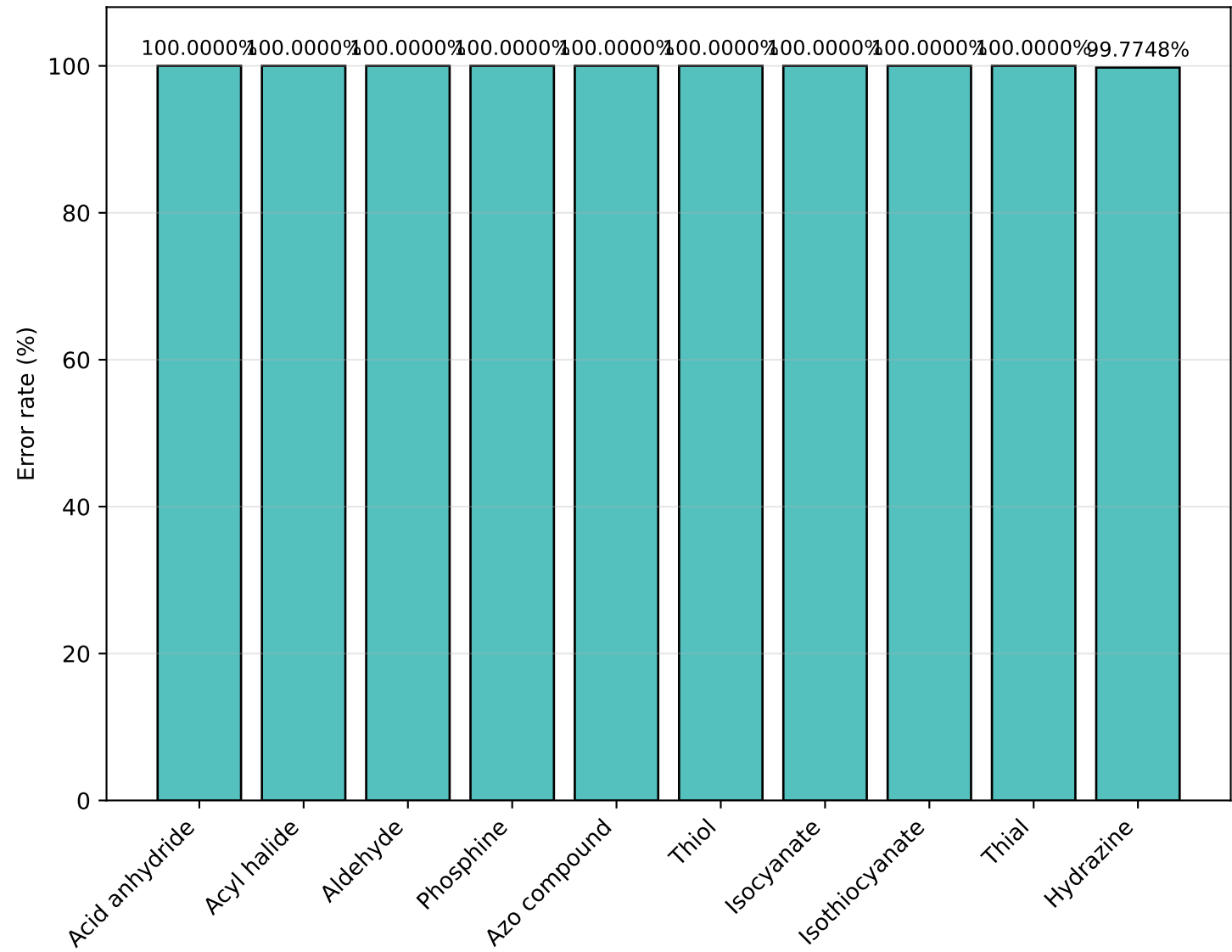
CNN MSMS- Validation Set - All 37 Functional Groups



Error-Rate Distribution



Top 10 Problem Groups



CNN MSMS- VALIDATION SET ERROR ANALYSIS

Global metrics:

F1 micro:	0.6688
F1 macro:	0.1228
Precision micro:	0.8012
Recall micro:	0.5740
Hamming accuracy:	0.9266

Error-rate statistics:

Mean:	83.5708%
Median:	96.4299%
Std:	29.8091%
Max:	100.0000% (Acid anhydride)
Min:	0.0000% (Sulfonic acid)

Problem classes:

Severe (>50%):	32 groups
High (20-50%):	2 groups
Moderate (5-20%):	0 groups
Low (<=5%):	3 groups